

20MCA281 - Internet of Things

Module 1

Module 1

Overview of Internet of Things:
Open-source semantic web
infrastructure for managing IOT
resources in the Cloud -
Device/Cloud Collaboration
framework for intelligence
applications

Internet of Things (IoT)

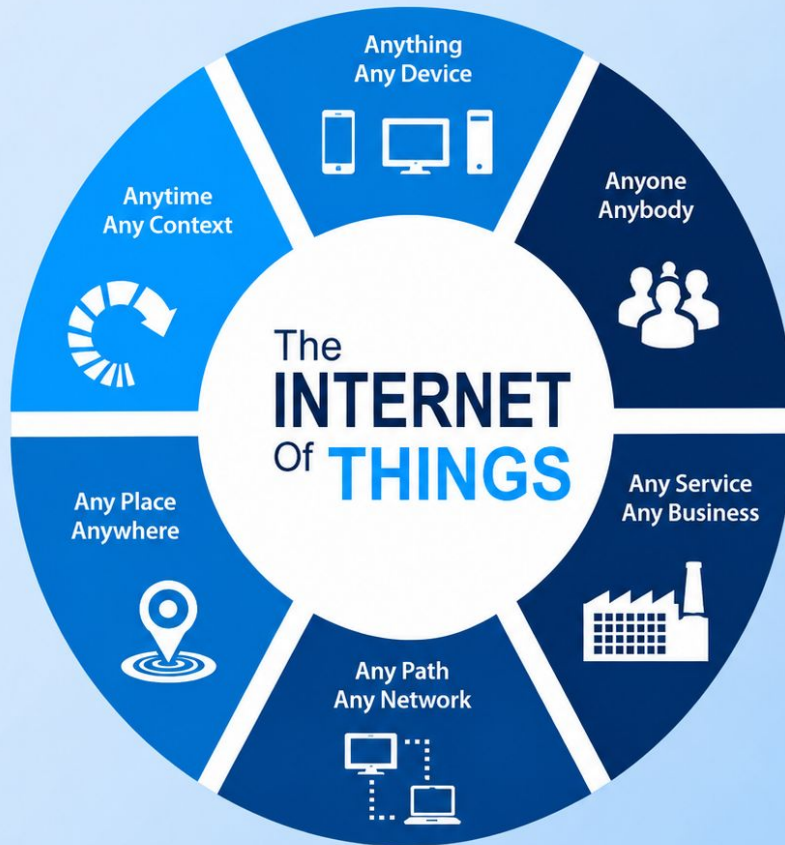
Introduction

The Internet has transformed the way people communicate. Earlier, the Internet mainly connected **people to people** through computers and mobile phones.

Today, technological advancements have enabled everyday physical objects to connect to the Internet. These objects can communicate, exchange information, make decisions, and perform tasks automatically without continuous human intervention.

This concept is known as the **Internet of Things (IoT)**.

IoT is considered one of the most significant technological revolutions after the Internet because it extends Internet connectivity beyond computers to billions of smart devices.



IoT

Connecting Things,
Transforming Lives

Definition of IoT

The Internet of Things (IoT) is a network of interconnected physical objects embedded with sensors, software, processors, and communication technologies that enable them to collect, exchange, process, and act upon data over the Internet with minimal human intervention.

IoT is a technology that connects everyday physical objects to the Internet so that they can sense, communicate, process information, and perform actions automatically.

Two important pillars of IoT: Internet and Things

IoT = Internet + Smart Things

1. Internet

Provides

- communication
- networking
- cloud connectivity
- remote access

Without the Internet, devices cannot exchange information globally.

2. Things

These are physical devices equipped with

- Sensors
- Controllers
- Communication modules
- Actuators

These objects collect information from the physical environment.

IoT = Internet + Smart Things

1. INTERNET

Provides



Communication



Networking



Cloud Connectivity

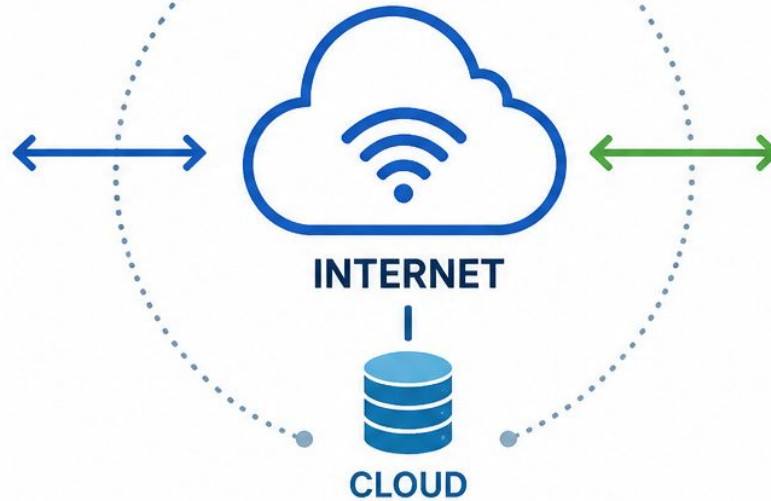


Remote Access

Without the Internet,
devices cannot exchange
information globally.

IoT

(Internet of Things)



2. THINGS

These are physical devices
equipped with



Sensors



Controllers



Communication
Modules



Actuators

These objects collect information
from the physical environment.



Smart Home



Healthcare



Industry



Transportation



Agriculture



Smart City

What is the Internet?

The **Internet** is a global network of interconnected computer systems that communicate using standard communication protocols such as TCP/IP.

It enables computers, mobile phones, servers, sensors, and other devices to exchange information regardless of their geographical location.

The Internet is often called the **"Network of Networks."**

Features of the Internet

- Global connectivity
- 24×7 availability
- Supports billions of users
- High-speed communication
- Resource sharing
- Data exchange
- Cloud connectivity

What are "Things"?

In IoT,

A **Thing** refers to any physical object that

- can sense its environment,
- can process information,
- can communicate over the Internet,
- and can perform actions automatically.

A thing does **not** have to be a computer.

It can be:

Car, Refrigerator, Fan, Camera, Traffic signal, Thermostat,
Medical equipment, Water pump, Street light, Factory machine

Why is IoT Needed?

Traditional systems rely heavily on manual monitoring and control, which can lead to delays, human errors, and inefficient resource utilization.

IoT addresses these challenges by enabling:

- Continuous monitoring
- Automatic data collection
- Real-time decision-making
- Remote control of devices
- Automation of repetitive tasks

This results in increased efficiency, reduced human effort, and improved quality of life.

Key Characteristics of IoT

An IoT system typically exhibits the following characteristics:

- **Connectivity:** Devices communicate through the Internet using protocols such as Wi-Fi, Bluetooth, ZigBee, or cellular networks.
- **Intelligence:** Embedded processors and software enable devices to make decisions.
- **Automation:** Devices perform tasks without continuous human intervention.
- **Context Awareness:** Devices respond based on environmental conditions such as temperature, location, or user activity.
- **Scalability:** IoT systems can support a large number of connected devices.
- **Interoperability:** Devices from different manufacturers can communicate using standard protocols.

- Every object capable of connecting to the Internet will fall into the “Things” category
- This notation is used to encompass a more generic set of entities, including smart devices, sensors, human beings, and any other object that is aware of its context and is able to communicate with other entities
- Making it accessible at any time, anywhere
- Objects are required to be accessible without any time or place restrictions.

- Initially, **Radio-Frequency Identification (RFID)** used to be the dominant technology behind IoT development
- **Wireless Sensor Networks (WSN) and Bluetooth-enabled devices are the IoT trend**
- Need to give more attention to the unique characteristics and requirements of IoT
- **Core Goal:** To make objects accessible **anytime and anywhere**, without time or place restrictions

- Focuses on connecting everyday physical objects such as devices, machines, and systems to computer networks
- Enables real-time data collection, monitoring, and automated actions
- Plays a key role in modern computer networks by extending Internet connectivity beyond traditional computers
- Improves efficiency, accuracy, and convenience in daily life and industrial applications

Components of IoT System

An IoT system is built using multiple interconnected components that work together to **sense data, process it, transmit it over networks, and present useful information to users or trigger actions automatically.**

Sensors collect data and send it via the network to edge or cloud systems where the data is processed and analyzed. Results are delivered to users through apps or used to automatically drive actuators and systems.

1. IoT Devices (Things)
2. Sensors and Actuators
3. Embedded Processing Units
4. Connectivity Modules
5. IoT Gateways
6. Cloud / Edge Computing
7. User Interface and Applications

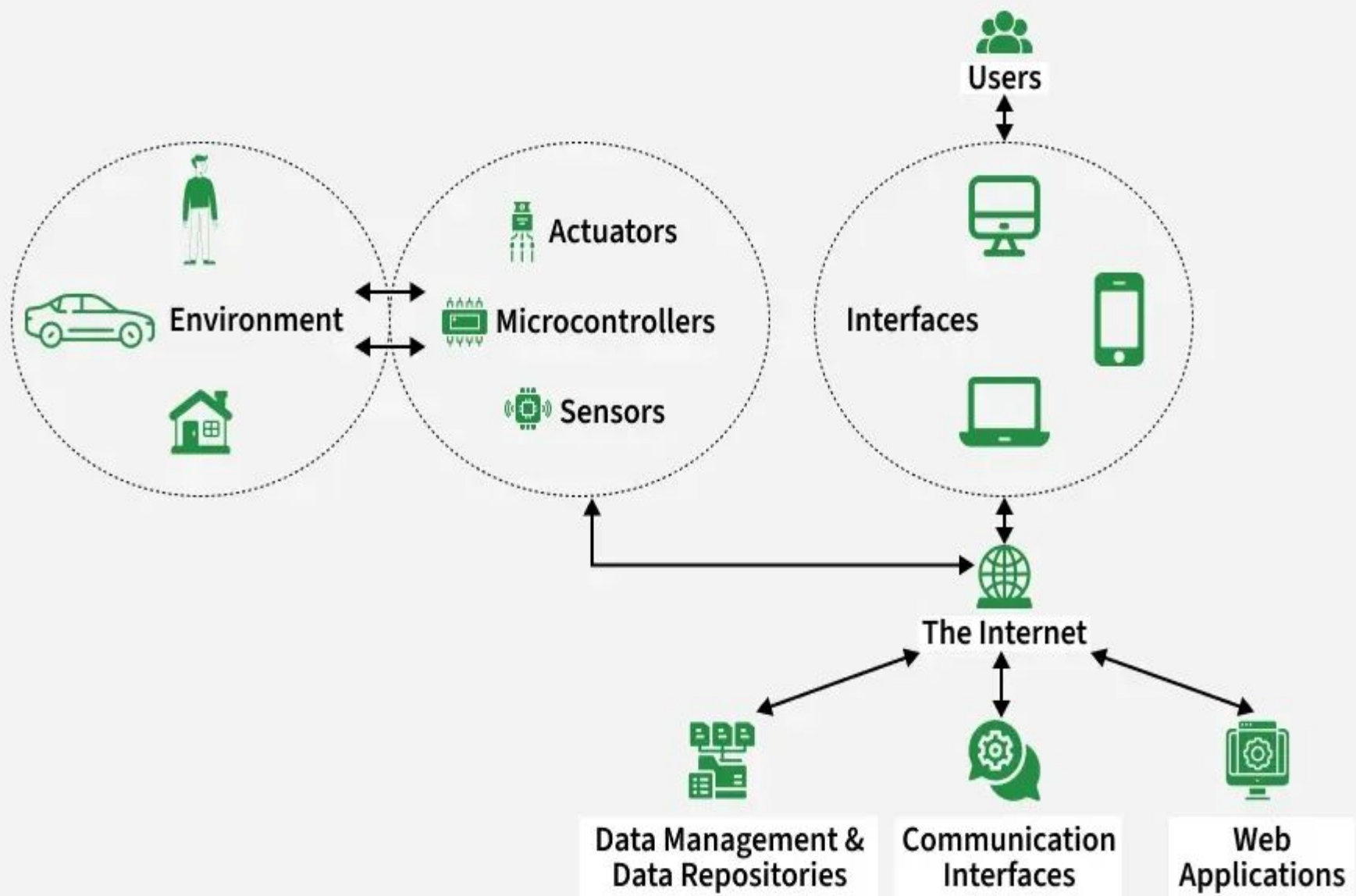
1. **IoT Devices (Things):** Physical objects such as appliances, machines, vehicles, or wearables that sense data from the environment or perform actions.
2. **Sensors and Actuators:** Sensors collect environmental data like temperature, motion, or light, while actuators convert digital signals into physical actions.
3. **Embedded Processing Units:** Microcontrollers or microprocessors that process sensor data, execute control logic, and manage device operations.
4. **Connectivity Modules:** Enable devices to communicate over networks using technologies such as Wi-Fi, Bluetooth, cellular networks, or LPWAN.

5. IoT Gateways: Aggregate data from multiple devices, perform protocol translation, and securely transmit data to cloud or edge platforms.

6. Cloud / Edge Computing: Cloud platforms store and analyze large IoT data, while edge computing processes data near the source to reduce latency.

7. User Interface and Applications: Dashboards, mobile apps, or web interfaces that allow users to monitor data, control devices, and receive alerts.

Note: Modern IoT systems also rely on low-power embedded systems, ensuring devices consume minimal energy while maintaining high performance.



IoT Definition Evolution

- **Internet Emergence** - Kevin Ashton is accredited for using the term “Internet of Things” for the first time during a presentation in 1999 on supply-chain management
- **Internet of Everything** - Since then, many definitions for IoT have been presented, that focuses mostly on connectivity and sensory requirements for entities involved in typical IoT environments

- New IoT definitions give more value to the need for ubiquitous and autonomous networks of objects.
- For example, Internet of Everything (IoE) is used by Cisco to refer to people, things, and places that can expose their services to other entities
- Industrial IoT (IIoT) - is another form of IoT applications favored by big high-tech companies.

- The fact that **machines can perform specific tasks** such as data acquisition and communication **more accurately** than humans has boosted IIoT's adoption
- **Machine to Machine (M2M) communication, Big Data analysis, and machine learning techniques** are major building blocks when it comes to the definition of IIoT
- **Smartness in IoT** - distinguishes IoT from similar concepts such as sensor networks
- Further categorized into **“object smartness”** and **“network smartness”**

- A smart network is a communication infrastructure characterized by the following functionalities:
 1. **Standardization and openness of the communication standards used**, from layers interfacing with the physical world (ie, tags and sensors), to the communication layers between nodes and with the Internet
 2. Object addressability and **multifunctionality**
 - (ie, the possibility that a network built for one application (eg, road-traffic monitoring) would be available for other purposes (eg, environmental-pollution monitoring or traffic safety))

- **Market Share** – Fast growing
- Market value of \$44.0 billion in 2011
- IoT market is expected to hit \$1423.09 billion by 2020
- Internet of Nano Things will play a key role in the future market and holding a value of approximately \$9.69 billion by 2020

Why do we need IoT

- **Scenario:** Patient at home is being monitored by an IoT application in the cloud
- Hospital is connected and monitoring the patient
- Any irregularity in patient health will alert hospital and patient can be taken to hospital
- Before patient reach hospital necessary arrangement can be done for treatment

- IoT is expanding the interdependence of humans to:
 - Interact
 - Contribute
 - Collaborate to things

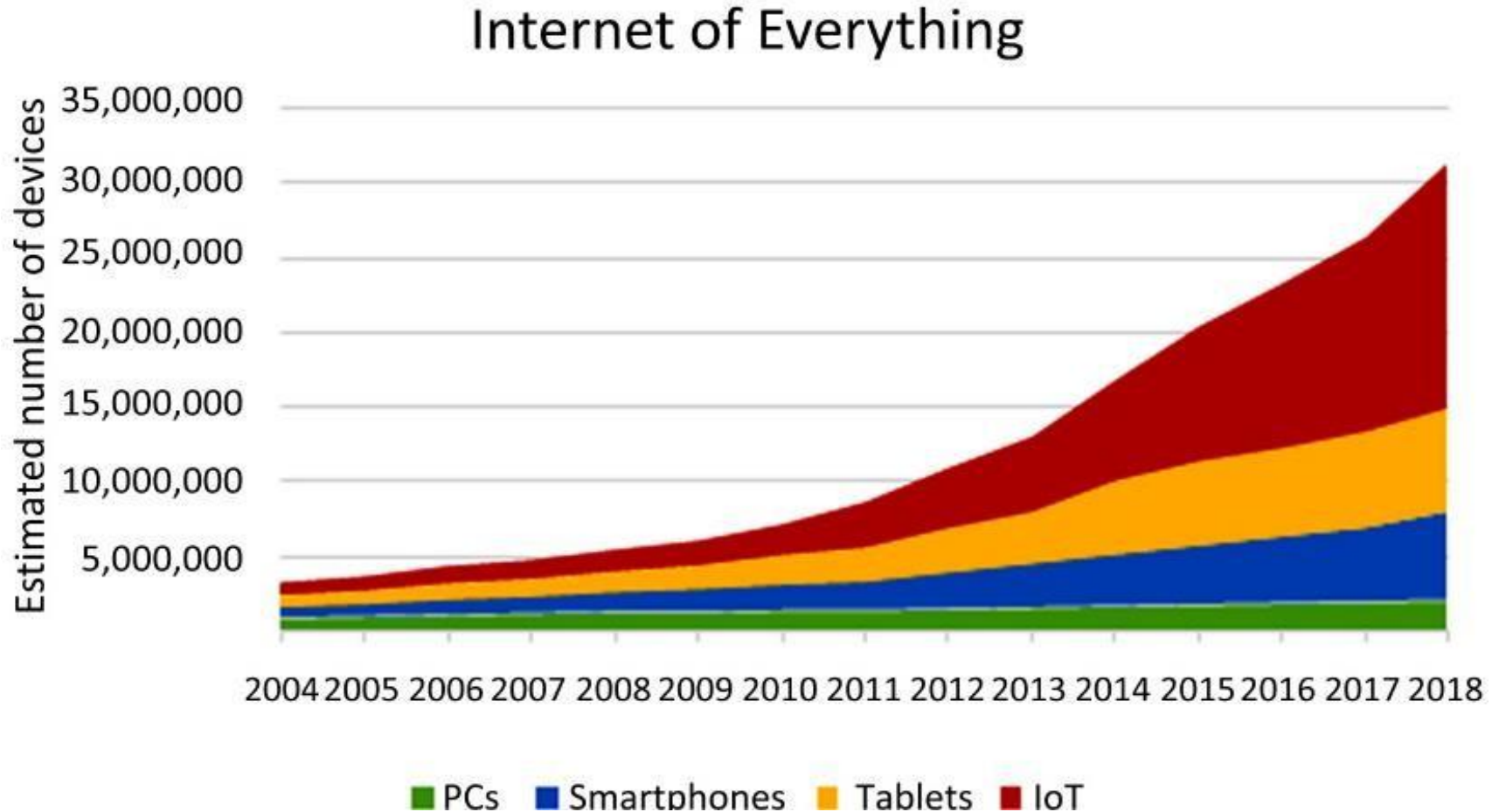
Benefits of IoT

- Efficient resource utilization
- Minimizing human effort
- Save time
- Development of AI through IoT
- Improved security

YouTube Videos – Basics of IoT

- <https://www.youtube.com/watch?v=6mBO2vqLv38>
- <https://www.youtube.com/watch?v=UrwbeOIlc68>
- <https://www.youtube.com/watch?v=h0gWfVC SGQQ>

IoT Trend Forecast



- **Human in the Loop** - IoT is also identified as an **enabler for machine-to-machine, human-to-machine, and human-with environment interactions**
- The trend of IoT is expected to shift toward the **fusion of smart and autonomous networks of Internet-capable objects**
- Advantages – Emergency management
- Healthcare

- **Improving the quality of life** - revolutionize the way our medical systems and businesses operate by:

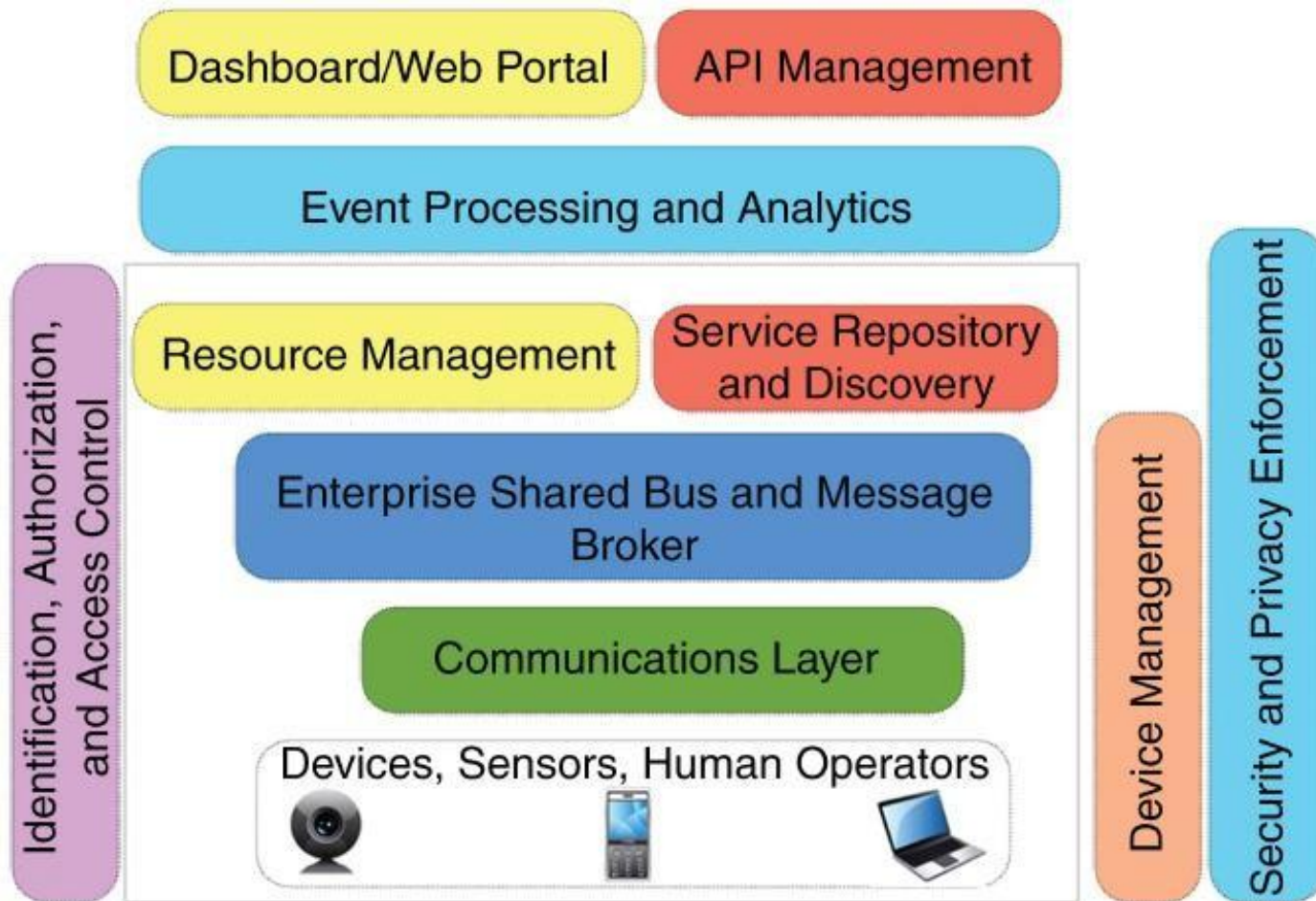
1. Expanding the communication channel between objects
2. Facilitating the automation and control process
3. Saving in the overall cost of implementation, deployment, and maintenance

IoT Architecture

- The building blocks of IoT are **sensory devices, remote service invocation, communication networks, and context-aware processing of events**
- IoT tries to picture a **unified network of smart objects and human beings responsible for operating them**

- A holistic system architecture for IoT needs to **guarantee flawless operation of its components** and link the physical and virtual realms together.
- Careful consideration is needed in designing failure recovery and scalability
- Need to have a certain level of adaptability to properly handle dynamic interactions within the whole ecosystem

Reference Architecture of IoT



- **Device Layer** – Various devices like sensors are interconnected
- Connected via Bluetooth, Zigbee, Wi-Fi etc.,
- **Communication Layer/Gateway Layer** – Supports REST (Representational State Transfer), MQTT (MQ Telemetry Transfer)/ HTTP
- **Bus/Aggregation Layer** – Acts as a message broker.
- Forms a bridge between the data and the communication layer for the sensors
- **Event Processing and Analytics Layer** – Provides ability to do web processing of data stored in database
- **Application Layer/API Management Layer** – Used to create web-based engines generates with external API's

- Different service and presentation layers are shown in this architecture
- Service layers include event processing and analytics, resource management and service discovery
- Also includes message aggregation and Enterprise Service Bus (ESB) services built on top of communication and physical layers
- API management for sharing system services and web-based dashboards are also included

Video

References

- <https://www.youtube.com/watch?v=KeaeuUcw02Q>

SOA Based Architecture

- Service Oriented Architecture
- Imperative for the service providers and users
- Ensures the interoperability among the heterogeneous devices
- Intensively used in WSN, due to its appropriate level of abstraction and advantages pertaining to its modular design
- Has the potential to augment the level of interoperability and scalability among the objects in IoT

- A generic SOA consists of four layers:
- **Sensing Layer**: It is integrated with available hardware objects **to sense the status of things**.
- **Network Layer**: It is the infrastructure to support over **wireless or wired connections** among 'things'
- **Service Layer**: It is to **create and manage services** required by users or applications.
- **Interface Layer**: It consists of the interaction methods with users or applications.

- From the user's perspective, **all services are abstracted into common sets, removing extra complexity for the user** to deal with different layers and protocols
- The ability to build diverse and complex services by composing different functions of the system (ie, modular composability) through service composition suits the heterogeneous nature of IoT
- **Accomplishing each task requires a series of service calls on all different entities spread across multiple locations**

API Oriented Architecture

- Conventional approaches like SOAP and RMI – creates overhead
- Web APIs and Representational State Transfer (REST)- based methods were introduced as promising alternative solutions
- The required resources range from network bandwidth to computational and storage capacity, and are triggered by request–response data conversions happening regularly during service calls

- Lightweight data-exchange formats like JSON can reduce the aforementioned overhead especially for smart devices and sensors with a limited amount of resources, **by replacing large XML files used to describe services**
- This helps in using the communication channel and processing the power of devices more efficiently
- Building APIs for IoT applications helps the service provider attract more customers while focusing on the functionality of their products rather than on presentation

IoT - Applications

- There are numerous IoT applications , which can be created to be specific to almost every industry.
- IoT applications are using Artificial Intelligence and Machine Learning to add intelligence to devices.
- Few applications are,
 - Smart Home
 - Health Care
 - Traffic Surveillance

Smart Home Applications

- Smart Home devices collect and share information with one another in an integrated platform and automate their actions based on the owner's preferences.
- **Smart Thermostats**: which monitor and control home temperatures to the comfort of owner.
- **Smart lighting**: lighting adjusts themselves based on the user preferences as well as external lighting.

- **Smart kitchen** : can check for smoke and carbon monoxide or temperature/humidity levels- special programs monitor if the users have enough products in the refrigerator, calculate nutritional value of the meals.
- **Security systems**: With special sensors, controllers can automatically lock the door, turn off electronic devices- users can check their home state remotely through app

Healthcare Applications

- **IoT for patients:** Devices in the form of wearable devices like fitness bands, blood pressure watches, ECG headbands etc.
- **IoT for physicians:** Data collected from the IoT devices can help physicians identify the best treatment process for the patients.
- **IoT for hospitals:** IoT devices tagged with the sensors are used for tracking real time location of medical equipments-Deployment of medical staff at different locations can also be analyzed real time.
- **IoT for health insurance companies:** The insurance companies can leverage data captured through health monitoring devices, which enable them to detect fraud claims

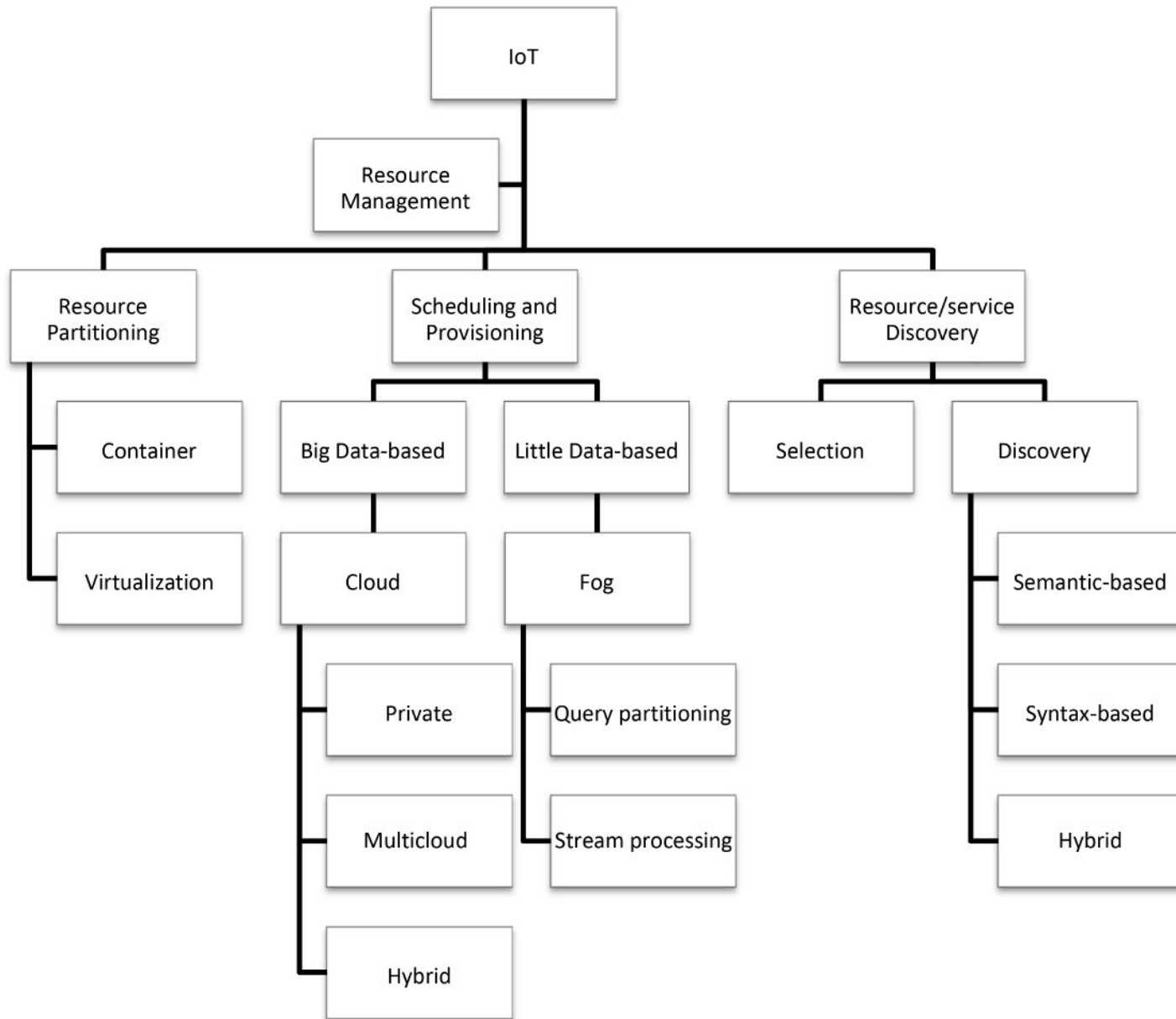
Traffic Surveillance

- Smart traffic control
- Vehicle control
- Smart parking
- Connected car
- Drowsiness alerts
- Electronic toll collection system
- Logistics and fleet management
and many others...

Resource Management

- Resource management involves discovering and identifying all available resources, partitioning them to maximize a utility function
- Which can be in terms of cost, energy, performance, etc., and, finally, scheduling the tasks on available physical resources

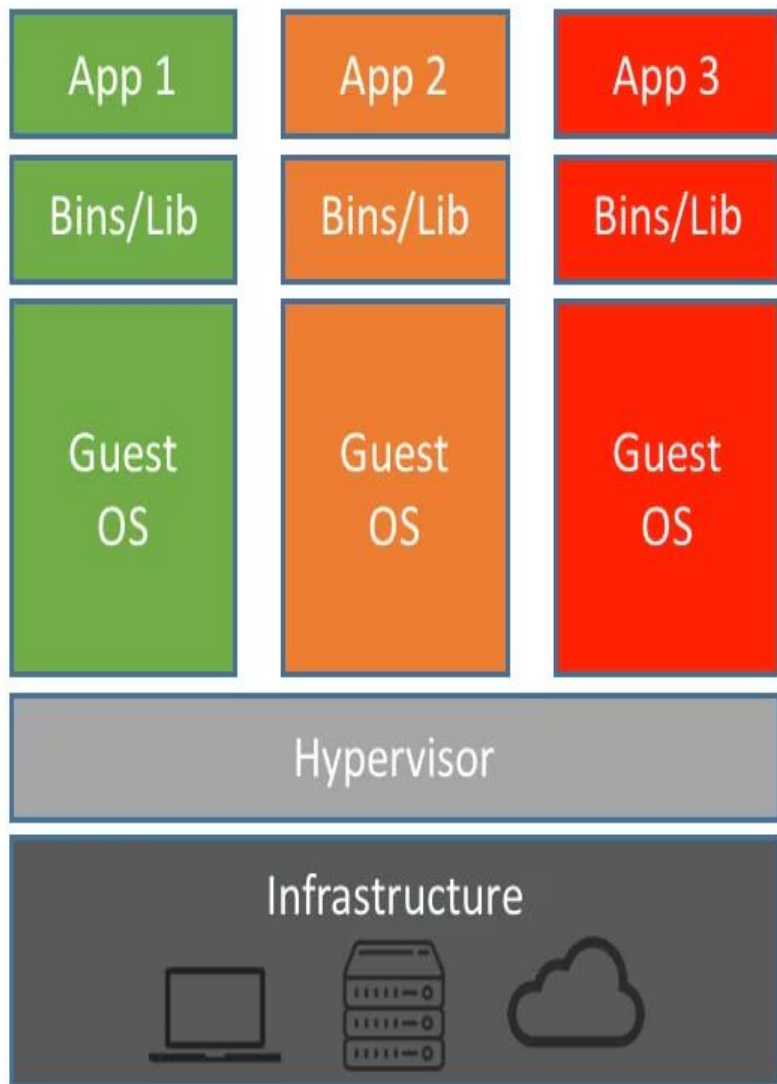
- Efficient resource management module needs
 - Robustness
 - Fault-tolerance
 - Scalability
 - Energy efficiency
 - QoS(Quality of Service)
 - SLA(Service Level Agreement)



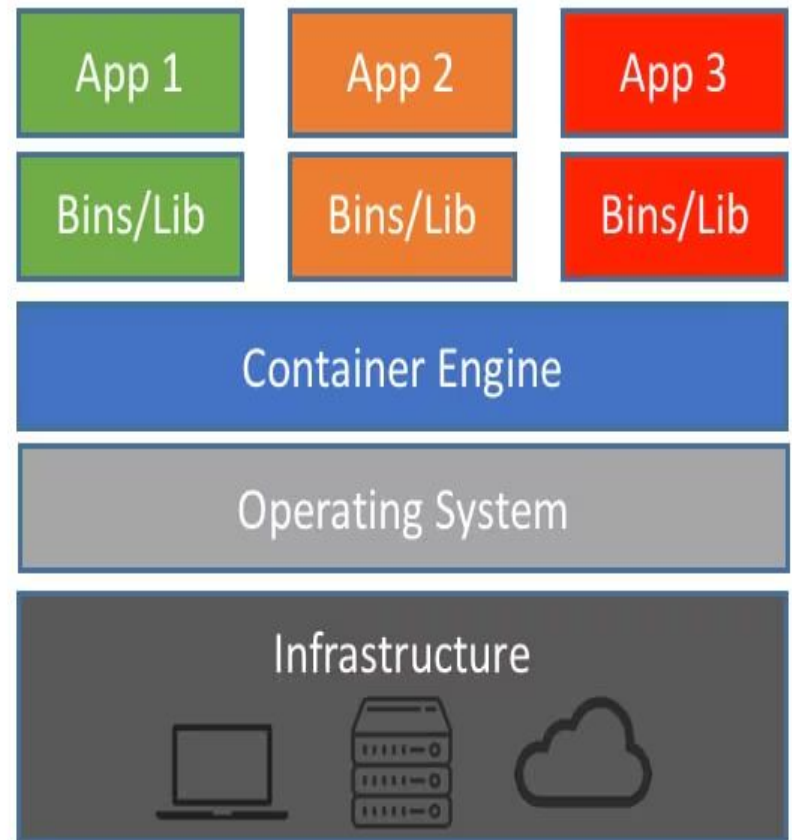
Resource Partitioning

- Partition the resources could gain a higher utilization rate
- Used in cloud computing via virtualization techniques and commodity infrastructures
- Since VMs, requires a considerable amount of memory and computational capacity, this configuration is not suitable for IoT

- To address these challenges, the concept of *Containers* has emerged as a new form of virtualization technology that can match the demand of devices with limited resources.
- Docker and Rocket are the two most famous container solutions.
- Containers are able to provide portable and platform-independent environments for hosting the applications and all their dependencies, configurations, and input/output settings



Machine Virtualization



Containers

Virtual Machines vs Containers

Feature	Virtual Machine	Container
Operating System	Separate Guest OS	Shares Host OS
Memory Usage	High	Low
Boot Time	Slow	Fast
Storage	Large	Small
Performance	Moderate	High
Resource Requirement	High	Low
Suitable for IoT	No	Yes

Computation Offloading

- **Computation Offloading** is the technique of moving heavy computational tasks from an IoT device to a more powerful remote computing system (such as the cloud or edge server) to improve performance and reduce the device's resource usage.
- Another solution for addressing the limitation of available resources in mobile and smart devices.
- The advantages of using code/computation offloading translate to more efficient power management, fewer storage requirements, and higher application performance

- Require the developers to manually add the functions required to execute on another device
- Using static code analyzers and dynamic code parsers is an alternative approach that results in better adaptivity in case of network fluctuations and increased latency

Identification and Resource/Service Discovery

- Current architectures proposed for creation of IoT environments lack support for an efficient and standard way of service discovery, composition, and their integration in a scalable manner
- The discovery module in IoT is twofold.
- The first objective is to identify and locate the actual device, which can be achieved by storing and indexing metadata information about each object.

- The final step is to discover the target service that needs to be invoked
- Lack of an effective discovery algorithm can result in execution delays, poor user experience, and runtime failures

IoT Data Management and Analytics

- Special considerations are required to process huge amounts of data originating from and circulating in, such a distributed and heterogeneous environment
- Big Data related procedures, such as data acquisition, filtering, transmission, and analysis have to be updated to match the requirements of the IoT

- Big Data is characterized by 3Vs, namely **velocity, volume, and variety**.
- This has led to the introduction of different data-processing approaches
- **Batch Processing and Stream Processing are two major methods**
- Lambda Architecture is an exemplary framework
- It has three main layers that enable the framework to support easy extensibility

- The first layer is called “Batch Layer” and hosts the master dataset and batch views where precomputed queries are stored
- Next is the “Serving Layer,” which adds dynamic query creation and execution to the batch views by indexing and storing them
- Finally, the “Speed Layer” captures and processes recent data for delay-sensitive queries

IoT and the Cloud

- Cloud computing, due to its on-demand processing and storage capabilities, can be used to analyze data generated by IoT objects in batch or stream format
- A pay-as-you-go model adopted by all cloud providers has reduced the price of computing, data storage, and data analysis, creating a streamlined process for building IoT applications

- With cloud's elasticity, distributed Stream Processing Engines (SPEs) can implement important features such as fault-tolerance and auto scaling for bursty workloads
- A framework has been proposed that supports sensor-data aggregation in cloud-based IoT context
- The framework is SOA-based and event-driven, and defines benefits from a semantic layer that is responsible for event processing and reasoning

- Most of the current design approaches for integrating cloud with IoT are based on a **three-tier architecture**:
- The bottom layer consists of IoT devices, middle layer is the cloud provider, and top layer hosts different applications and high- level protocols
- Limits the practicality and full utilization of cloud computing in scenarios

Real-Time Analytics and Fog Computing

- Current data-analytics approaches mainly focus on dealing with Big Data
- Processing data generated from millions of sensors and devices in real time is more challenging
- Proposed solutions cannot address the latency constraints of real-time applications

- Real-time processing requirements and the increase in computational power of edge devices such as routers, switches, and access points lead to the emergence of the **Edge Computing paradigm**
- The **Edge layer contains** the devices that are in closer vicinity to the end user than the application servers, and can include **smartphones, smart TVs, network routers**, and so forth

- Processing and storage capability of these devices can be utilized to extend the advantages of using cloud computing by creating another cloud, known as **Edge Cloud**
- **Decrease networking delays, save processing or storage cost, perform data aggregation, and prevent sensitive data from leaving the local network**

- Fog Computing is an extension of cloud computing that aims to keep the same features of Cloud, such as networking, computation, virtualization, and storage
- It also meets the requirements of applications that demand low latency, specific QoS requirements, Service Level Agreement (SLA) considerations, or any combination of these
- These extensions can ease application development for mobile applications, Geo-distributed applications such as WSN, and large-scale systems used for monitoring and controlling other systems

Cloud Versus Fog		
	Fog	Cloud
Response time	Low	High
Availability	Low	High
Security level	Medium to hard	Easy to medium
Service focus	Edge devices	Network/enterprise core services
Cost for each device	Low	High
Dominant architecture	Distributed	Central/distributed
Main content generator—consumer	Smart devices—humans and devices	Humans—end devices

Cloud vs. edge vs. fog computing

	Cloud computing	Edge computing	Fog computing
LOCATION	Centralized in remote data centers	Close to data creation and consumption points	Hierarchical architecture between edge devices and cloud
PROCESSING POWER	Highest processing capabilities	Limited processing power	More powerful than edge devices but less than cloud
SCALABILITY	Highly scalable	Less scalable	More scalable than edge computing
ARCHITECTURE	Centralized model	Decentralized architecture	Hierarchical architecture with multilevel connections
LATENCY	Higher latency due to distance	Low latency for real-time processing	Reduced latency compared to cloud
SECURITY	Advanced security measures	Security focused on local devices	Security measures for edge devices
USE CASES	Long-term, in-depth analysis and storage	Real-time processing (autonomous computing, AR/VR)	Complex analytics and data processing across network
NODE DISTRIBUTION	Centralized	Millions of nodes far from cloud	Billions of nodes closer to cloud
DATA HANDLING	Stores and processes all data	Processes data directly on devices	Analyzes data importance before sending to cloud
DEPLOYMENT MODEL	Often multi-tenant	Localized deployments	Distributed across network

A comparison between layers

Cloud Computing

- Centralized
- High processing & computing
- High latency
- AI processing power
- Highest storing capacity
- Cybersecured

Fog Computing

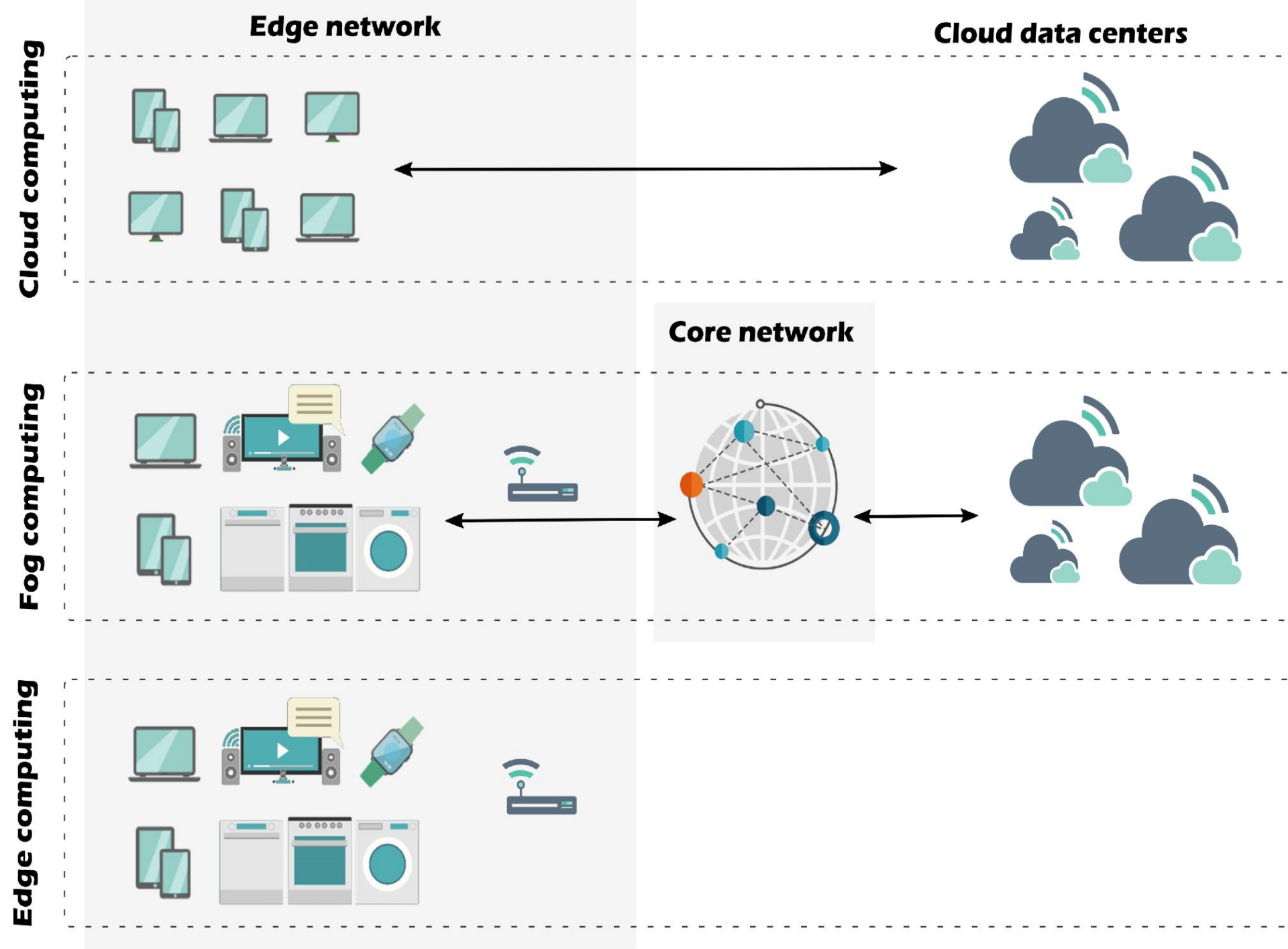
- Decentralized
- Low latency
- Dedicated bandwidth where most needed
- AI processing power
- Networking effect with other edge site nodes
- Cybersecured

Edge Computing

- Decentralized
- Lowest latency
- Saves Bandwidth
- Processing & networking limitations
- Potential data privacy issues
- Cybersecurity concerns

IoT Devices

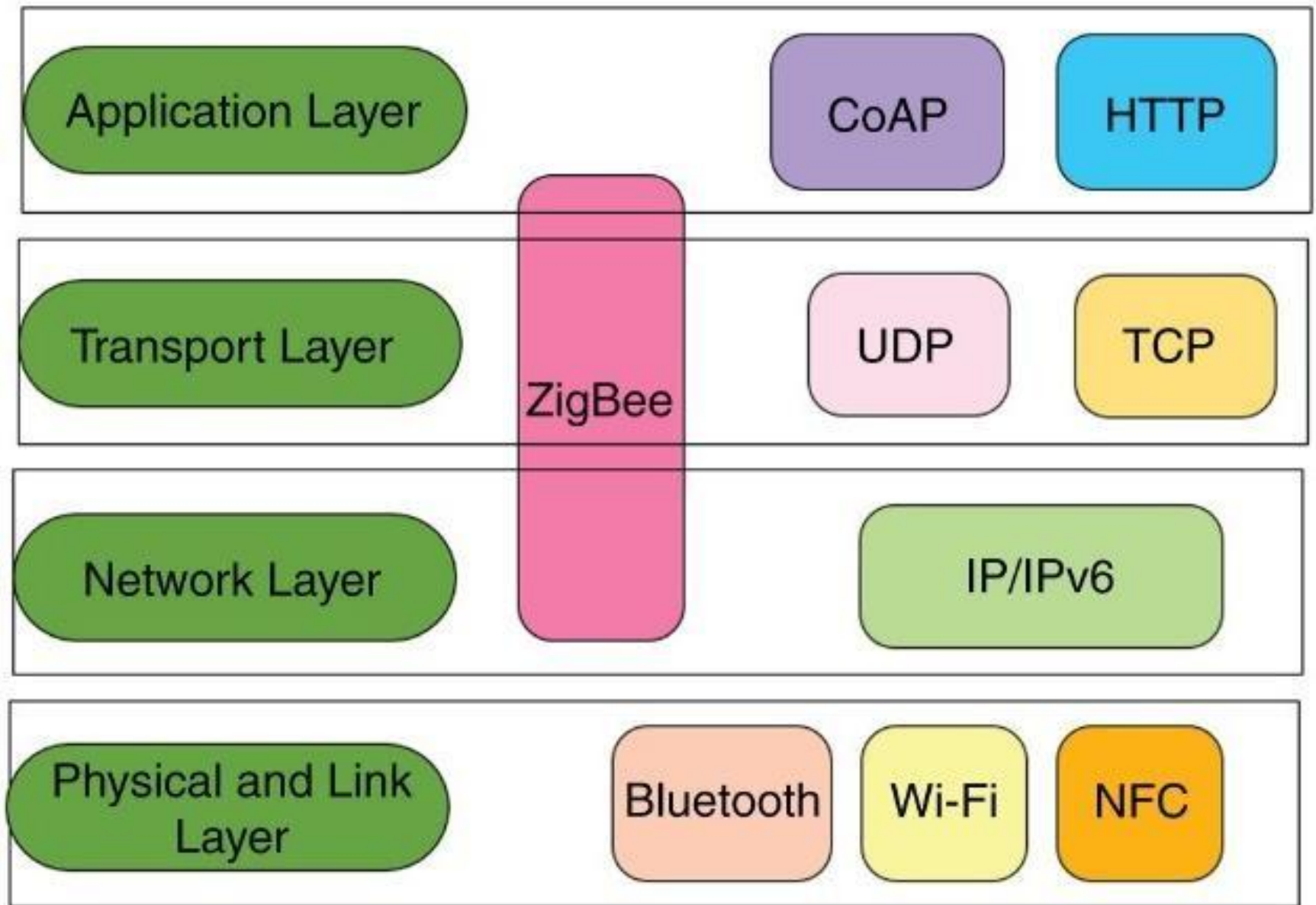
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Communication Protocols

- IoT can be viewed as an aggregation of different networks, including mobile networks (3G, 4G, CDMA, etc.), WLANs, WSN, and Mobile Adhoc Networks (MANET)
- Seamless connectivity is a key requirement for IoT
- Network-communication speed, reliability, and connection durability will impact the overall IoT experience

- With the emergence of high-speed mobile networks like 5G, and the higher availability of local and urban network communication protocols such as Wi-Fi, Bluetooth, and WiMax, creating an interconnected network of objects seems feasible, however, dealing with different communication protocols that link these environments is still challenging



Physical and Link Layer

- Is the connecting medium
- Determines how data is sent physically over the connecting medium
- Twister pair, Optical fibre or wireless mode
- Wired/Wireless
- Eg., IEEE 802.3 (Ethernet), 802.11 (Wi-Fi), 802.16 (WiMAX) 2G/3G/4G

Network Layer

- Based on the device's specification (memory, CPU, storage, battery life), the communication means and protocols vary.
- However, the commonly used communication protocols and standards are listed below:

- **RFID** (eg, ISO 18000 series that comes with five classes and two generations, and covers both active and passive RFID tags)
- **IEEE 802.11** (WLAN), IEEE 802.15.4 (ZigBee), Near Field Communication (NFC), IEEE 802.15.1 (Bluetooth)
- Low-power Wireless Personal Area Networks (**6LoWPAN**) standards by IETF
- M2M protocols such as MQTT and CoAP
- IP layer technologies, such as IPv4, IPv6, etc.

Transport and Application Layer

- Protocols in IoT can be classified into three categories:
 1. General-purpose protocols like IP and SNMP that have been around for many years and are vastly used to manage, monitor, configure network devices, and establish communication links

2. Lightweight protocols such as CoAP that have been developed to meet the requirements of constrained devices with tiny hardware and limited resources
3. Device or vendor-specific protocols and APIs that usually require a certain build environment and toolset

IoT Applications

- IoT promises an interconnected network of uniquely identifiable smart objects
- This infrastructure creates the necessary backbone for many interesting applications that require seamless connectivity and addressability between their components
- The range of IoT application domain is wide and encapsulates applications from home automation to more sophisticated environments, such as smart cities and e-government

- Monitoring:
 - Monitoring devices via APIs can be helpful in multiple domains.
 - The APIs can report power usage, equipment performance, and sensor status, and they can perform actions upon sending predefined commands

1. Smart Home: mainly target energy-saving and monitoring

- Support different communication protocols (Wi-Fi, Bluetooth, etc.)

2. **Smart metering:** can help in identifying production or performance defects via application of anomaly detection on the collected data, and thus increase the productivity

3. **Buildings:** helps to move toward green solutions, save energy, and, consequently, minimize operation cost

4. **Healthcare**

IoT devices monitor patients continuously by collecting health data such as:

Heart rate, Blood pressure, Blood glucose level, Body temperature

This enables remote patient monitoring and faster medical intervention.

- Business Process and Data Analytics:
 - *Society level*, where IoT mainly influences and improves government services by reducing cost and increasing government transparency and accountability
 - *Industry level*, in which manufacturing, emergency services, retailing, and education have been studied as examples
 - *Organizational level*, in which IoT can bring the same type of benefits as those mentioned in society level

- *Individual level*, where daily life improvements, individual efficiency, and productivity growth are marked as IoT benefits
- Information gathering and collaborative consumption:
 - Social Internet of Things (SIoT) is where IoT meets social networks
 - SIoT concept, motivated by famous social media like Face- book and Twitter, has the potential to affect many people's lifestyles

IOT Applications and Usecases



Security

- As adoption of IoT continues to grow, attackers and malicious users are shifting their target from servers to end devices
- In terms of physical accessibility, smart devices and sensors are far less protected than servers
- Having physical access to a device gives the attackers an advantage to penetrate with less hassle

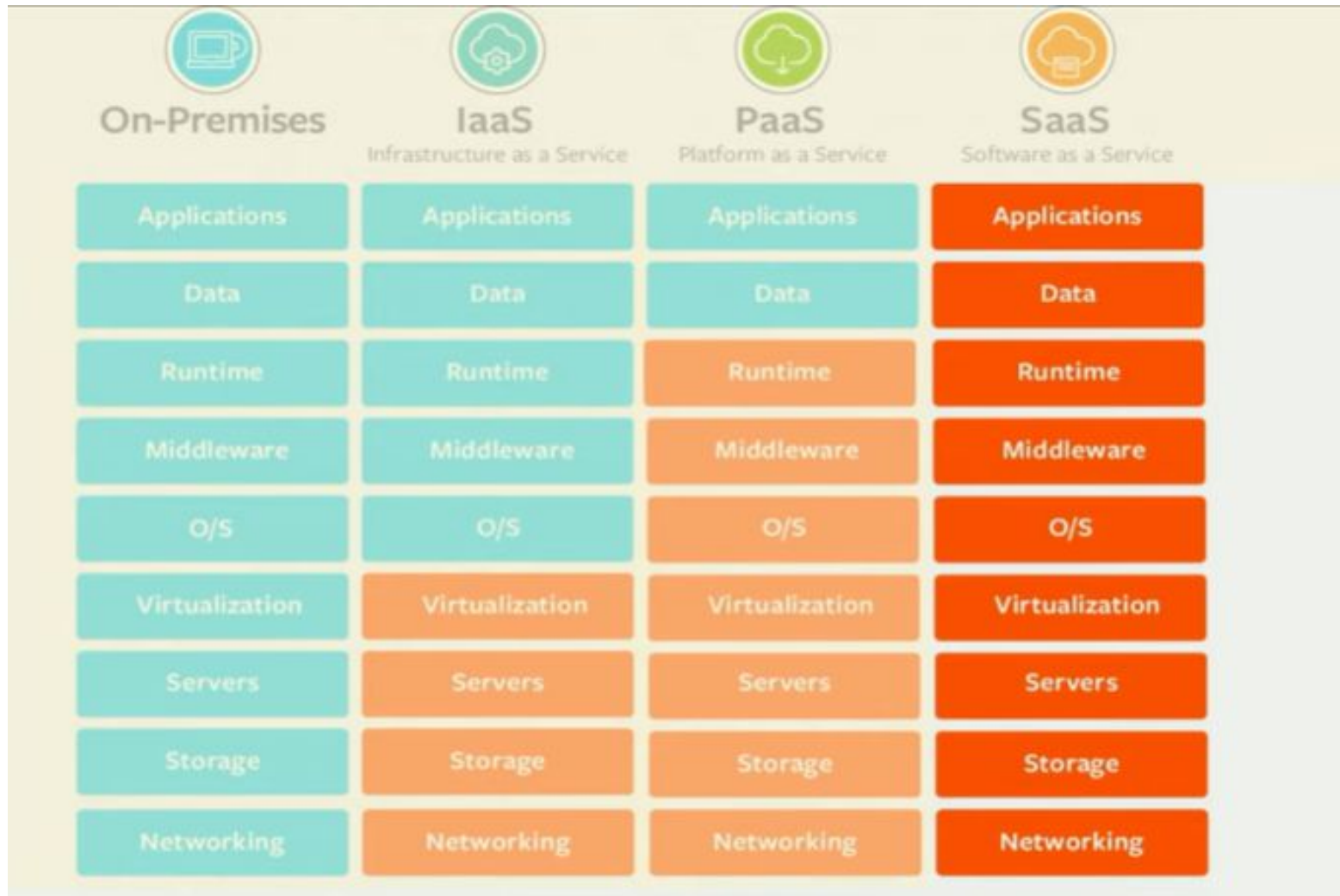
- The number of devices that can be compromised are far more than the number of servers
- Since devices are closer to the users, security leads to leaking of valuable information and has catastrophic consequences
- Due to heterogeneity and the distributed nature of IoT, the patching process is more time consuming, thus opening the door for attackers

Open Source Semantic Web Infrastructure For Managing IoT Resources in the Cloud

- Cloud computing and internet of things are nowadays two of the most prominent and popular paradigms that are expected to shape the next era of computing
- Depending on the type of computing resources delivered via the cloud, cloud services take different forms

- Infrastructure as a service (IaaS), Platform as a service (PaaS), Software as a service (SaaS), Storage as a service (STaaS), and more
- These services hold to promise to deliver increased reliability, security, high availability, and improved QoS at an overall lower total cost of ownership

- Blue color - handling by user
- Orange color - handling by cloud



Infrastructure-as-a-Service (IaaS)

Infrastructure as a Service is the most flexible category of cloud services. It aims to give you complete control over the hardware that runs your application



Purchase, installation, configuration, and management of their own software operating systems, middleware, and applications.



Responsible for ensuring that the underlying cloud infrastructure (such as virtual machines, storage, and networking) is available for the user.

Platform-as-a-Service (PaaS)

PaaS provides an environment for building, testing, and deploying software applications. The goal of PaaS is to help you create an application quickly without managing the underlying infrastructure.



Responsible for the development of their own applications.



Responsible for operating system management, and network and service configuration.

Software-as-a-Service (SaaS)

SaaS is software that is centrally hosted and managed for the end customer. It is usually based on an architecture where one version of the application is used for all customers, and licensed through a monthly or annual subscription.

Examples of SaaS software:

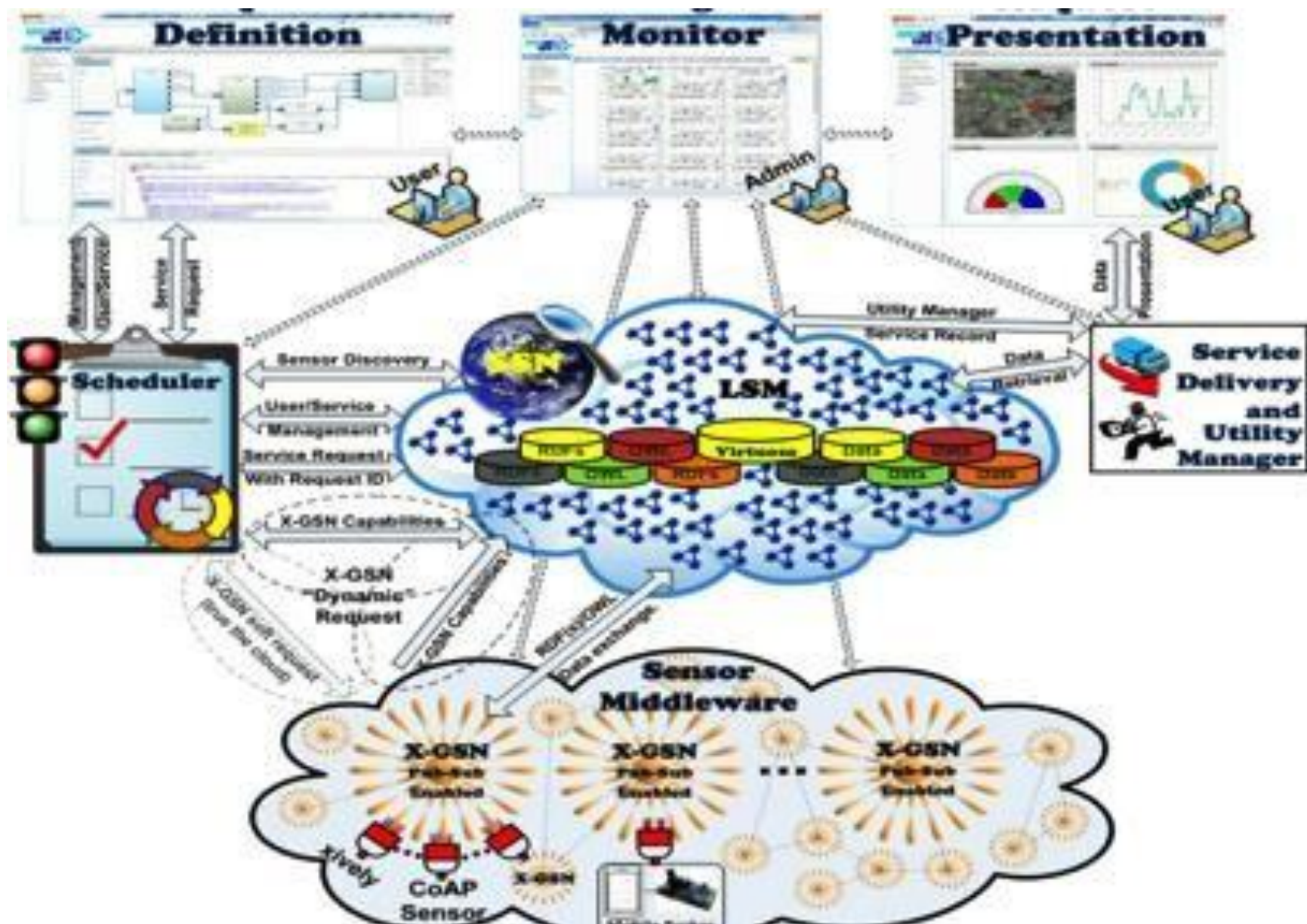
- Office 365
- Skype.
- Dynamics CRM Online.



Users just use the application software; they are not responsible for any maintenance or management of that software.



The cloud provider is responsible for the provision, management, and maintenance of the application software.



OPEN IoT ARCHITECTURE FOR IoT/CLOUD CONVERGENCE

- **1. Sensor Middleware** acts as a **bridge (hub)** between the physical world and the OpenIoT platform **which collects, filters, and combines data streams stemming from virtual sensors or physical-sensing devices**. Transfers processed data to the cloud.
- Eg, (**virtual**) signal-processing algorithms, information- fusion algorithms, and **social-media data streams**
- Eg, (**physical**) **temperature sensors, humidity sensors and weather stations**

- This middleware acts as a hub between the OpenIoT platform and the physical world
- It enables the access to information stemming from the real world
- Deployed on the basis of one or more distributed instances which may belong to different administrative entities
- *The Cloud Computing Infrastructure*, which enables the storage of data streams stemming from the sensor middleware, thereby acting as a cloud database

- Also stores metadata for the various services, as part of the scheduling process
- **2. Cloud Computing Infrastructure**
 - Computational (software) components of the platform could also be deployed in the cloud in order to benefit from its elasticity scalability, and performance characteristics
 - The cloud infrastructure could be either a public infrastructure (Amazon Elastic Compute Cloud (EC2)) or a private infrastructure (such as a private cloud deployed, based on Open Stack)

- The cloud infrastructure can be characterized as a sensor cloud, given that it primarily supports storage and management of sensor data-streams
- **3. Directory Service**, which stores information about all the sensors that are available in the OpenIoT platform
- It also provides the means for registering sensors with the directory, as well as for the look-up (discovery) of sensors

- The IoT/cloud architecture specifies the use of semantically annotated descriptions of sensors as part of its directory service
- The Directory Service is deployed within the cloud infrastructure, thereby providing the means for accessing sensor data and metadata residing in the cloud
- **4. Global Scheduler**, which processes all requests for on-demand deployment of services and ensures their proper access to the resources that they require

- Do the task of parsing the service request and discover the sensor that can contribute to its fulfillment
- It also performs reservation of resources
- **5. Local Scheduler component**, executed at the Sensor Middleware level
- Ensures optimized access to the resources managed by sensor middleware instances
- Global scheduler regulates the access to the resources of the OpenIoT platform (data stream in the cloud)

- ***The Service Delivery and Utility Manager*** , performs a dual role

1. It combines the data stream within the OpenIoT system to deliver the requested service
 - Makes use of the service description and resources identified and reserved by the (Global) scheduler component
2. Act as a service-metering facility which keeps track of utility metrics for each individual service

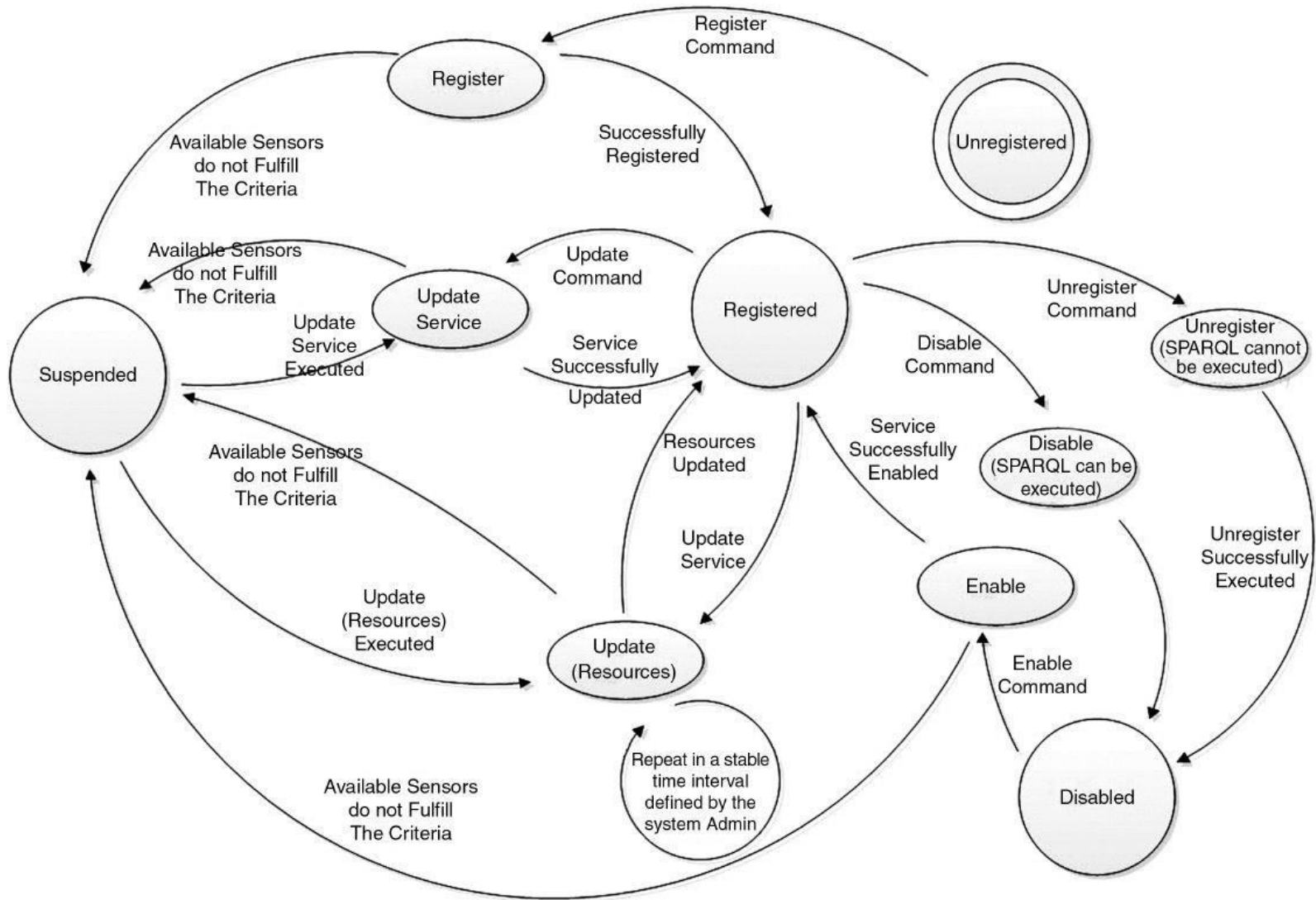
- **7. Request Definition tool** , enables the specification of service requests to the OpenIoT platform
- It comprises a set of services for specifying and formulating such requests while also submitting them to the global scheduler
- Features a Graphical User Interface (GUI)
- **8. Request Presentation component** , in charge of the visualization of the outputs of an IoT service

- Selects mashups from an appropriate library in order to facilitate service presentation
- ***The Configuration and Monitoring component***, enables management and configuration functionalities over the sensors, and the IoT services which are deployed with in the platform
- This platform is supported by a GUI

SCHEDULING PROCESS AND IoT SERVICES LIFECYCLE

- The Global scheduler component is the first entry point for service requests
- Performs **two main functions** towards the delivery of the service:
 1. Selection of the sensors involved
 2. Reservation of the needed resources
- Scheduler **manages all the meta data** of the IoT service including:

1. The signature of the service (i/p and o/p)
2. Sensors used to deliver the service
3. Execution parameters associated with the service (eg. intervals in which the service shall be repeated, other resources used etc.,)
 - Global scheduler component keep track of and controls the life cycle of IoT services
 - The following life cycle management services are supported by the scheduler



State Diagram of the OpenIoT Services Lifecycle Within the Scheduler Module

- ***Resource Discovery :***
 - Discovers a virtual sensor's availability
 - Provides the resources that match the requirements of a given request for an IoT service
- ***Register:***
 - Responsible for establishing the requested service within the cloud database
 - Initially identifies the log of all sensors which are needed for delivering the requested IoT service

- Metadata of the IoT services are persisted to the cloud
- ***Unregister:***
- The resource allocated for services (eg., sensors) are released
- In the case of an active service, a deactivation process is initially applied
- Status of the service is updated in the data structures holding the metadata about the service

- ***Suspend:*** Service is deactivated and the operation is ceased.
- As part of suspension the platform does not release the resources associated with the service.
- ***Enable from suspension:*** Enables a previously suspended service.
- The data structure holding the service's meta data in the cloud are appropriately updated

- ***Enable***: Allows the enablement of an unregistered service
- ***Update***: Permits changes in the IoT service
- Allows for the updating of service's lifecycle metadata according to the requested changes
- ***Registered service status***: Provides the life cycle status of a given IoT service
- Metadata about IoT service is provided
- ***Service Update Resources***: Periodically checks all enabled services.

- ***Get service***: retrieves the description of a registered service
- ***Get available services***: returns a list of registered services that are associated with a particular user

Scheduling and Resource Management

- The OpenIoT scheduler enables the availability and provision of accurate information about the data requested by each service as well as about the sensors that will be used in order to deliver these data
- A wide range of different resource management and optimization algorithms can be implemented at the scheduler component of the OpenIoT architecture

- The scheduler components maintain information that could optimize the use of the **network** on the basis of aggregate operations such as the **aggregation of queries** to the sensor cloud
- A variety of **in-network processing** and data management techniques can be implemented in order to **optimize processing times** and/or **reduce the required access to the sensor network**

- The criteria for aggregating queries and their results could be based on common spatial regions
- Another class of optimizations are caching techniques
- It reduce network traffic, enhance the availability of data to the users (sink), and reduce the cost of expensive cloud-access operations
- Involves maintaining sensor (data streams) data to a cache memory in order to facilitate fast and easy access to them

Device/Cloud Collaboration framework for intelligence applications

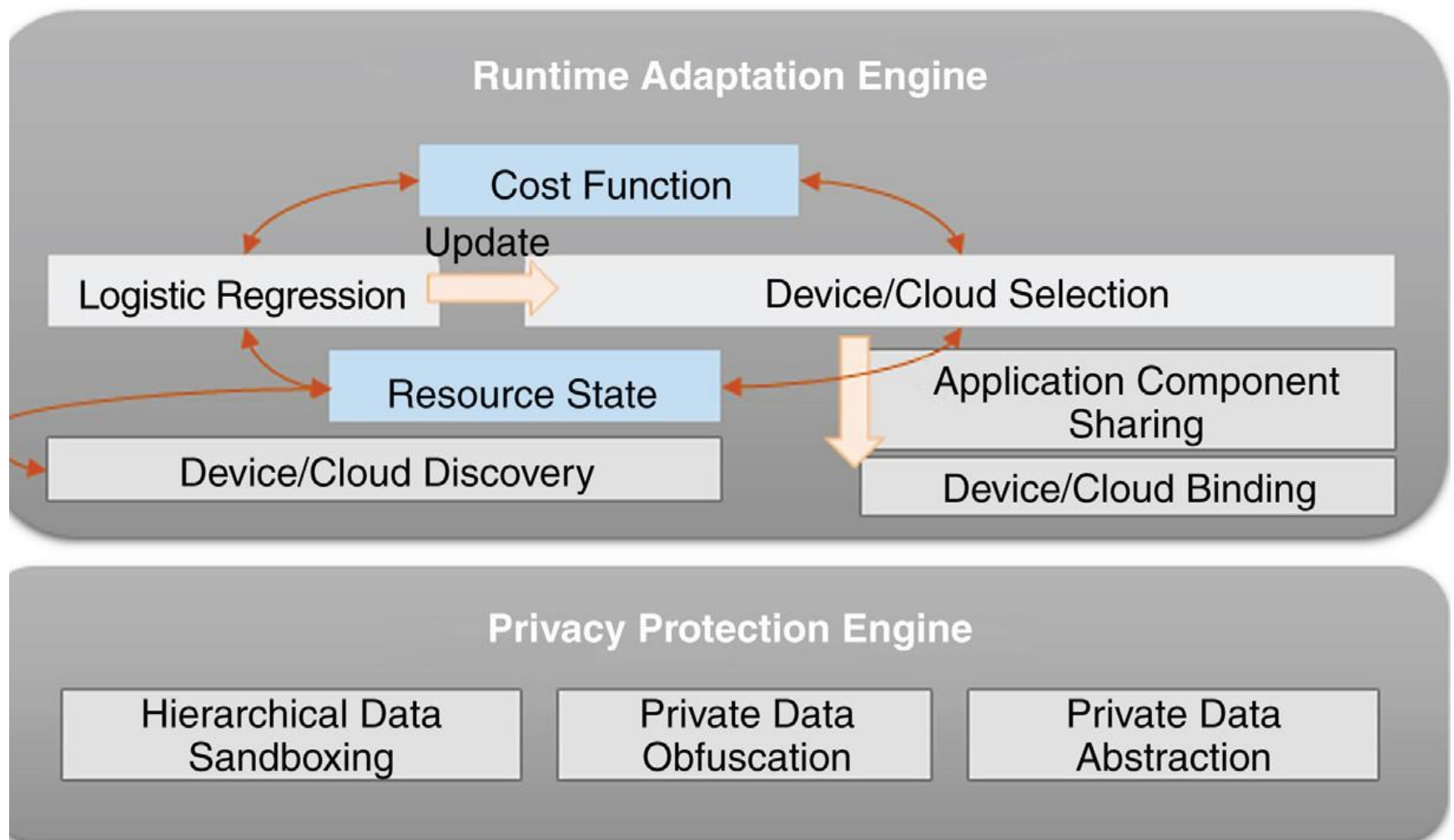
- Cloud offers on-demand computing and storage resources to the users who cannot afford the expenditure on computing equipment and management workforce
- This led to the notable commercial success of Amazon's EC2 and Microsoft's Azure
- These companies have adopted the business model of renting out their virtualized resources to the public

- Google runs a popular proactive service, **Google Now**, which gives individualized recommendations based on the user's context inferred from the personal data
- Facebook leverages the user-uploaded images and social-network data to automatically recognize users and the relationship among them

Device/Cloud Collaboration Framework

- **Powerful smart mobile devices:**
- Smartphones have enough computing capacity to process various computing tasks.
- The device usage can be constrained by limited battery life and network connectivity.
- Consider utilizing the highly available cloud resources in addition to the device.

Device/Cloud Collaboration Framework



- **Runtime adaptation engine:**
- Given a computation task, our framework **first faces a problem of choosing the entity to execute the task**
- Suppose we are given a task of processing a query issued over voice using a voice-query processing engine embedded in a smartphone
- **Mobile engine answer the query without the cost of transferring it to the cloud**
- **Consumes the limited battery life, and the accuracy of the result may not be good.**

- The user may have to issue the query redundantly to the cloud-based query processing engine with the hope of getting a better result
- This may hurt the overall Quality of Experience (QoE)
- This calls for a decision mechanism that automatically selects a better *agent* that can execute a given task
- With automatic selection process user do not have to worry about going through extra interaction cycles for determining where to run a job

- **Runtime Adaptation Engine (RAE)** sits at the **core of the framework**.
- The RAE maintains **a list of available devices and cloud** to utilize Device/Cloud Discovery, and **monitors the state of their available resources**.
- The RAE employs a **Logistic Regression algorithm** to learn the **most cost-effective policy for distributing tasks** among devices and cloud, given the resource state.

- Definition of the cost function is the **weighted sum of the resource state** (battery life, network, and CPU usage...)
- The policy obtained by running the Logistic Regression is enforced by the **Device/Cloud Selection module** that **chooses the most economical compute resources**, based on the expected cost-value for a given task.
- The mechanism of the **RAE is an autonomous agent**, which can be deployed on each device and cloud.
- **RAEs communicate with each other** to transparently share the resource state **for determining the ways to distribute a given workload**.

- The cloud-side RAE can also model its own cost function as a **weighted sum of residual CPU cycles and storage space** across the entire infrastructure.
- If the residual capacity falls below particular thresholds, the cloud may have to reject the resource-sharing request coming from the paired devices. This is because running the requested task would be too costly.
- The cloud-side RAE advises the device-side RAE to **either execute the task within the device or simply wait for the compute resource on the cloud to be freed up.**
- The RAEs on the devices and the cloud make decisions autonomously, without any supporting brokerage system in the middle.

- Device-side RAE has the burden of **periodically monitoring the state of the cloud resources.**
- But ,the cloud-side RAE **does not have to monitor the resource state of the millions of paired devices**, as the cloud makes a relatively simple decision, that is, to either reject or accept a task-execution request.
- By default, **this framework does not consider offloading cloud-initiated tasks to the devices.**

- ***Device/Cloud Discovery component:***
- To support task distribution among the neighboring devices, the device/cloud collaboration framework implements discovery of devices
- ***Application Component Sharing :***
- Suppose device A wants to collaborate with its neighboring device B which is not equipped with appropriate application components to process a given task. Then, device A can transfer the necessary application components to device-B through the Application Component Sharing

Applications of Device Cloud Collaboration

- Context-aware proactive suggestion
- **Semantic QA caching**
- Automatic image/speech recognition